

NET621 –Weekly Practical Week 1

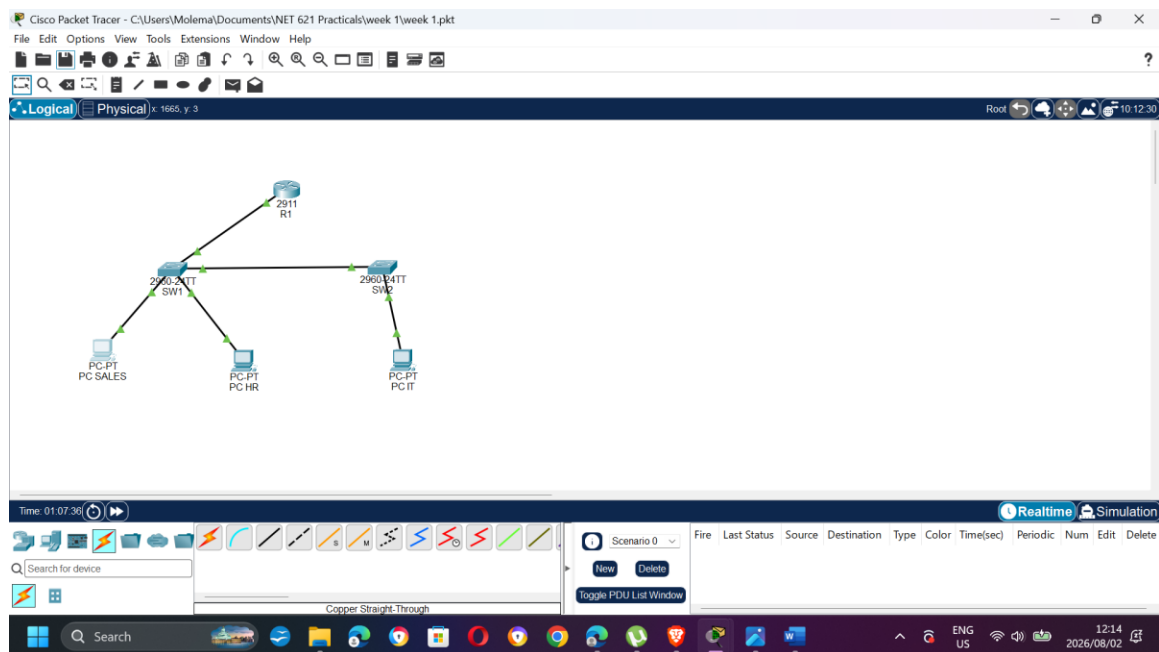
VLANs, Trunking & Inter-VLAN Routing – Implementation Report

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This report documents the successful configuration of VLANs, trunk links, and Router-on-a-Stick inter-VLAN routing as per Week 1 tasks.



1. Devices Used

- 1x Router: Cisco 2911 (R1)
- 2x Switches: Cisco 2960-24TT (SW1 and SW2)
- 3x End Devices: PC-PT (PC SALES, PC HR, PC IT)
- Cables: Copper Straight-Through (PC to Switch, Switch to Switch, Switch to Router)
- Topology: SW1 Fa0/1 → PC SALES, SW1 Fa0/2 → PC HR, SW2 Fa0/1 → PC IT, SW1 Fa0/24 ↔ SW2 Fa0/24 (Trunk), SW1 Fa0/3 ↔ R1 G0/0 (Trunk to Router)

2. Screenshot of Topology (Matches Diagram)

Figure 1 shows the logical topology created in Cisco Packet Tracer with all devices interconnected.

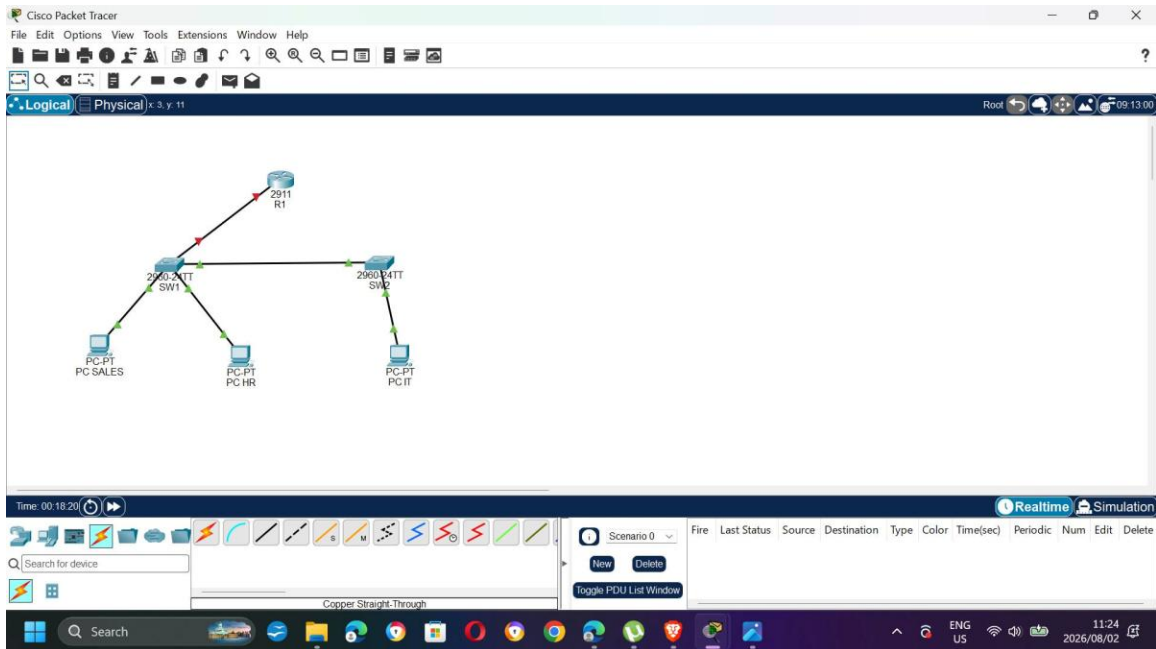


Figure 1: Full topology – R1 connected to SW1, SW1 trunked to SW2, PCs connected to access ports. All links green.

3. Step 1: Create VLANs on SW1 and SW2

Commands executed on both switches:

```
vlan 10
name SALES
vlan 20
name HR
vlan 30
name IT
exit
```

Screenshot of SW1 VLAN creation:


```
switchport mode access
switchport access vlan 20 (HR)
```

SW2:

```
interface fa0/1
switchport mode access
switchport access vlan 30 (IT)
```

5. Step 3: Configure Trunk Links

On SW1:

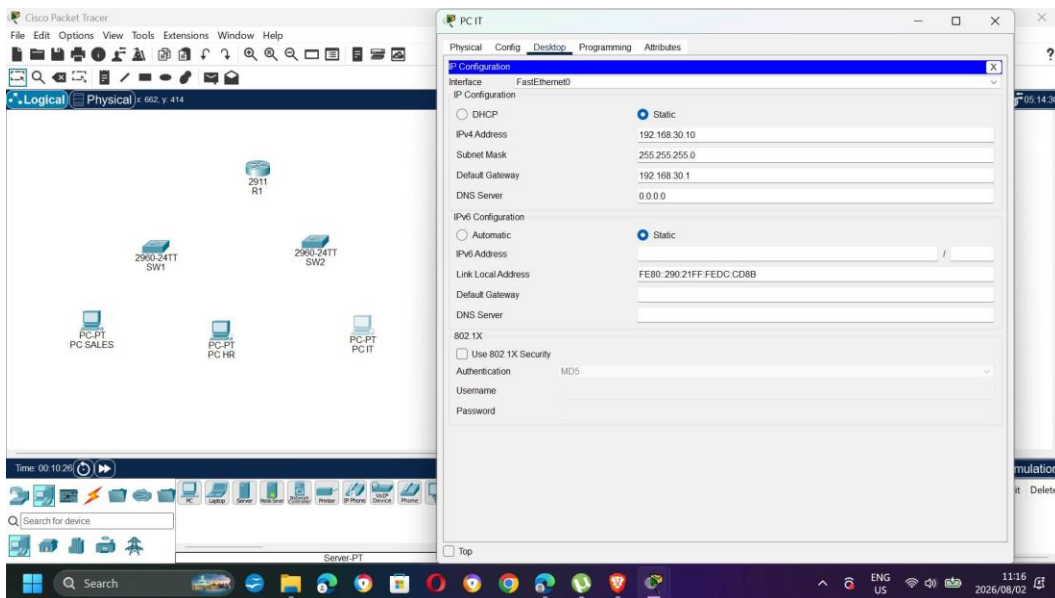
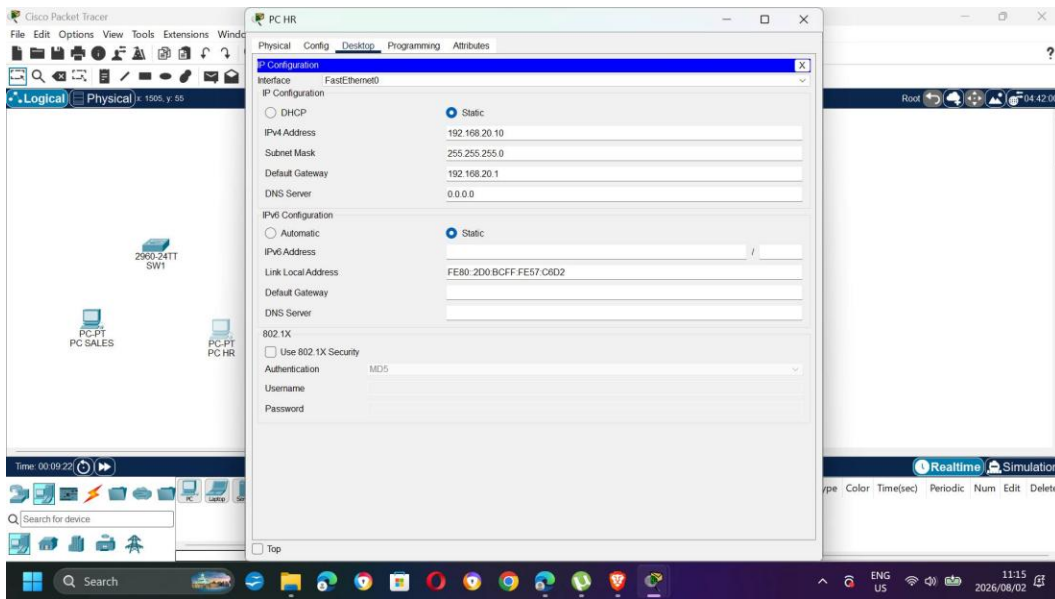
```
interface fa0/3
switchport mode trunk (Link to R1)
interface fa0/24
switchport mode trunk (Link SW1-SW2)
```

On SW2:

```
interface fa0/24
switchport mode trunk
```

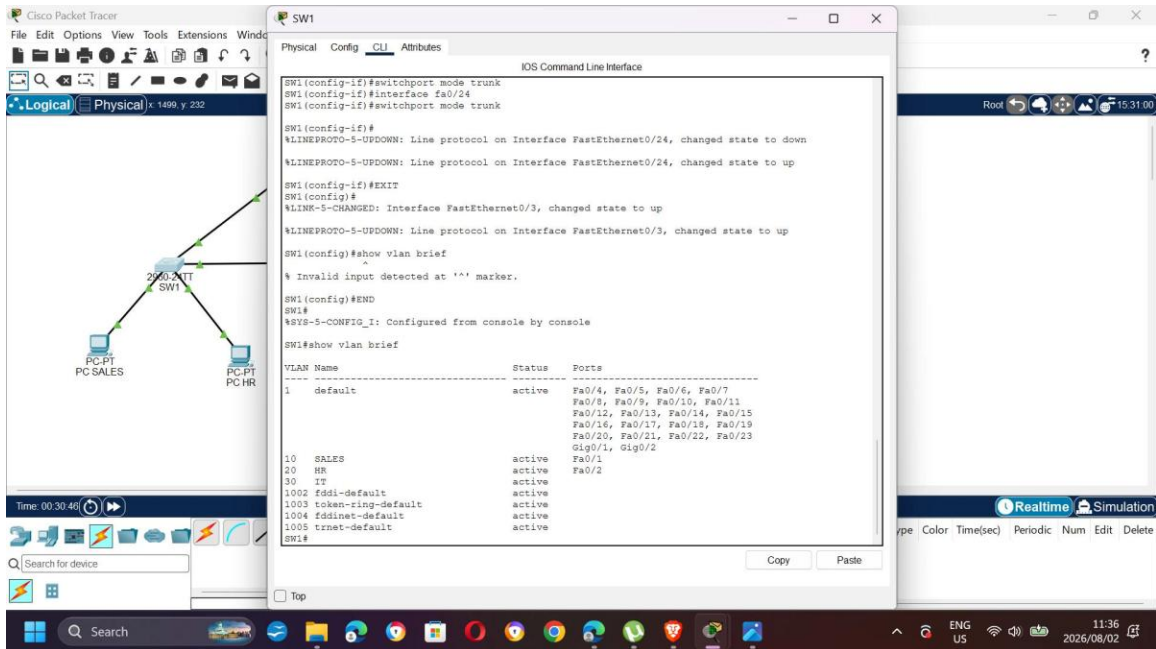
6. Step 4: Configure Router-on-a-Stick on R1

```
interface g0/0
no shutdown
interface g0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
interface g0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
interface g0/0.30
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
```

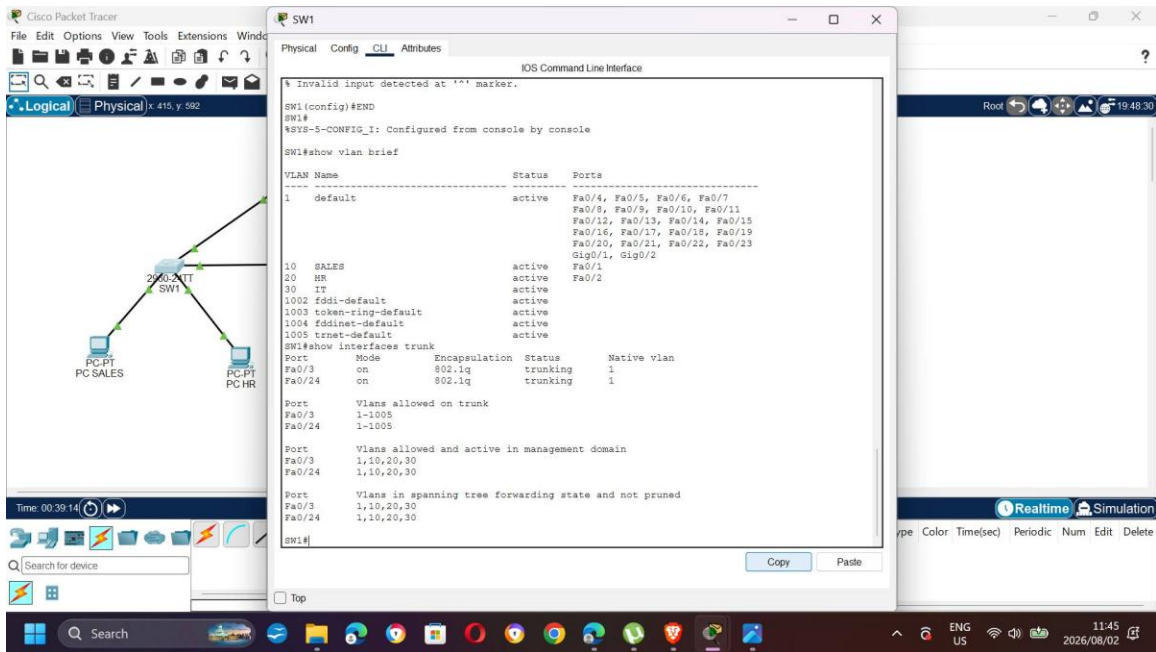



8. Verification – show vlan brief and show interfaces trunk

Output of show vlan brief on SW1 shows VLANs 10,20,30 active with Fa0/1 in VLAN 10 and Fa0/2 in VLAN 20:

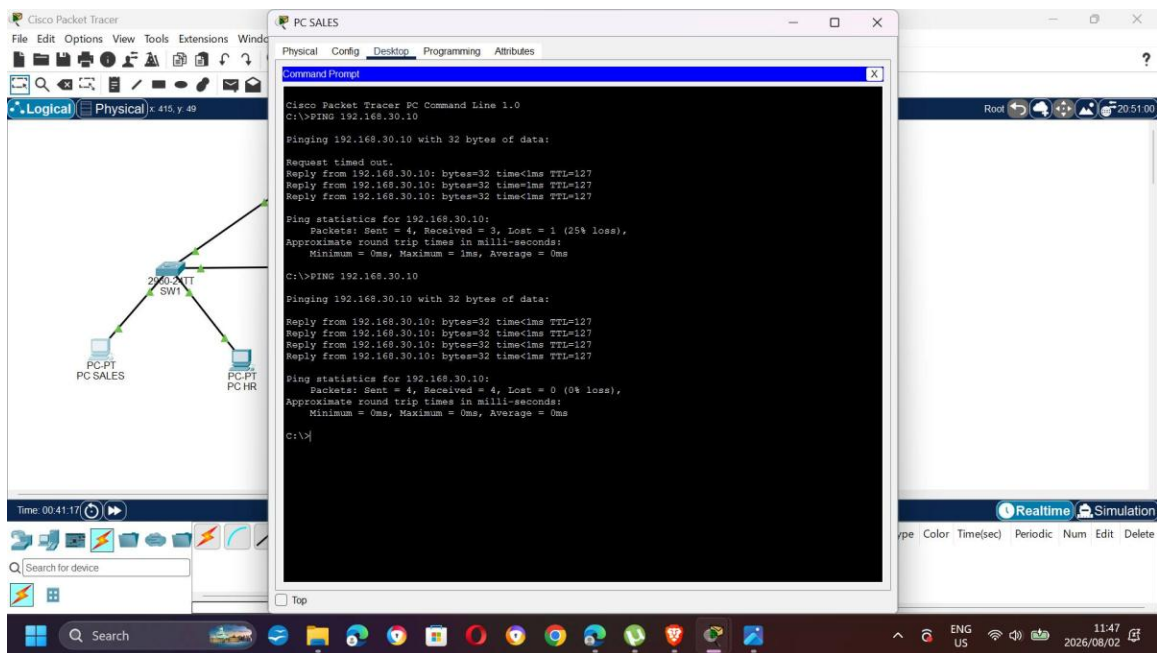


Output of show interfaces trunk on SW1 shows Fa0/3 and Fa0/24 in trunking mode, encapsulation 802.1Q, allowing VLANs 1,10,20,30:



9. Verification – Cross-VLAN Ping

Successful ping from PC SALES (192.168.10.10) to PC IT (192.168.30.10) proves R1 is routing between VLANs:



Result: Packets Sent=4 Received=4 Lost=0 (0% loss) – Inter-VLAN routing working. First ping timed out due to ARP, subsequent pings <1ms TTL=127.